

Synthetic Market Phase 7

Relation-invariance follow-through on the relational representation report. This phase keeps the anchor-pair relation fixed while changing style, layout, roster-rank context, and global magnitude scale.

DATE	PROMPTS	DESIGN	OBJECT
21 March 2026	384 market-only	4 scenarios × 2 styles × 4 layouts × 4 rosters × 3 scales	anchor-pair relation

MAIN READ

Phase 7 confirms the relation-first shift: anchor-pair market relations survive layout, roster-rank, and magnitude changes.

This is the first synthetic phase that cleanly answers the “what next after row retrieval?” question. The corrected analysis shows that relation identity beats both rank-bucket matching and scale-bucket matching, with uniformly strong margins. The caveat is that the whole task still peaks at `row_mean @ layer 1`, so the dataset is likely too easy to reveal deeper structure.

BEST PRIMITIVE R ²	BEST RELATION	BEST RANK CTRL	BEST SCALE CTRL
0.99998	0.2679	0.2677	0.2312
row_mean L1	roster-only, momentum edge	row_mean L1	row_mean L1

Why Phase 7 Exists

The previous relational report established the right object shift:

- row retrieval is too brittle
- pairwise relations are a better upstream target

Phase 7 executes the next logical step from that report:

- keep the same anchor-pair relation
- move that relation through different roster-rank contexts
- rescale the whole market magnitude
- and ask whether the relation vector still groups by relation identity rather than by rank or scale alone

This is a stronger test than Phase 6 because it adds two explicit nuisance axes:

- **roster-only**: same style/layout/scale, different rank context

- **magnitude-only**: same style/layout/roster, different absolute scale

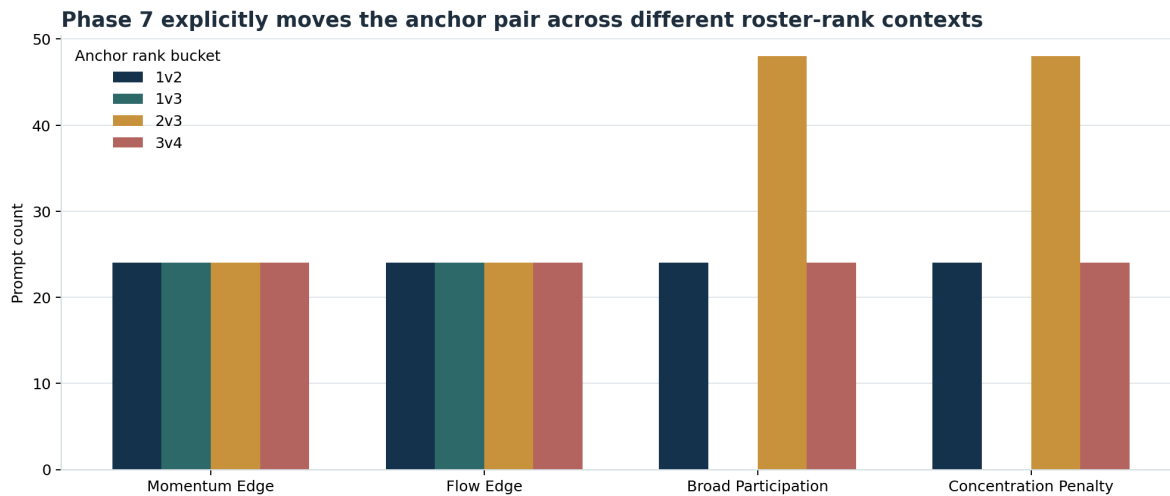
Primitive Factors Are Still Explicit

Primitive-factor decode remains effectively perfect in this family.

Variable	Best State	Metric	Layer
pct_5m	row_mean	R ² 0.99998	L1
net_flow_5m	row_mean	R ² 0.99998	L1
unique_traders_5m	row_mean	R ² 0.99998	L1
top20_holder_pct	row_mean	R ² 0.99997	L1
attractiveness_score	row_mean	R ² 0.99998	L1
risk_adjusted_score	row_mean	R ² 0.99998	L1

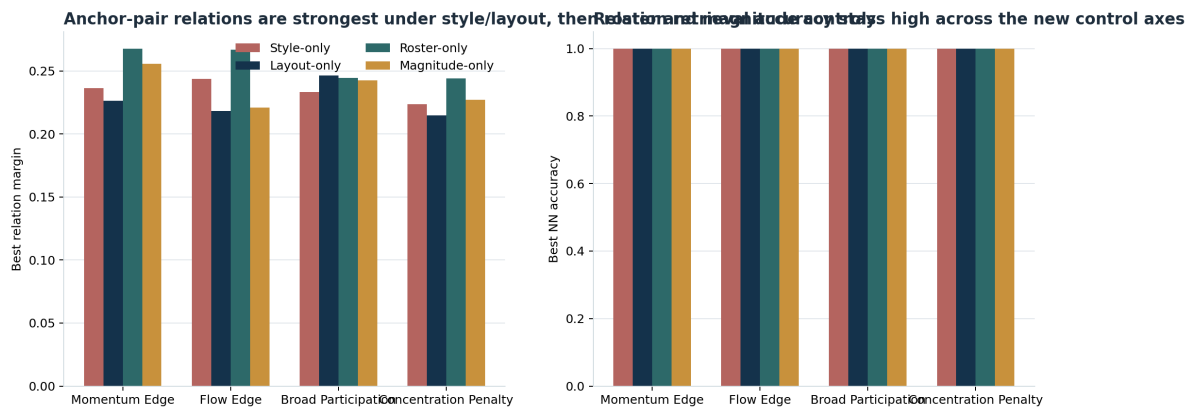
That matters for interpretation: Phase 7 is not discovering that the row holds market information. It is testing whether the **comparative object** built from those rows stays stable across stronger nuisance changes.

The Rank Context Is Real



The anchor pair is intentionally moved across different roster-rank contexts. This is not a cosmetic reformatting test; the same latent relation is evaluated under new relative position in the market.

Relation Identity Survives The New Axes



Best relation margins and nearest-neighbor accuracy across control modes. The signal stays strong not only under style and layout changes, but also under roster-only and magnitude-only controls.

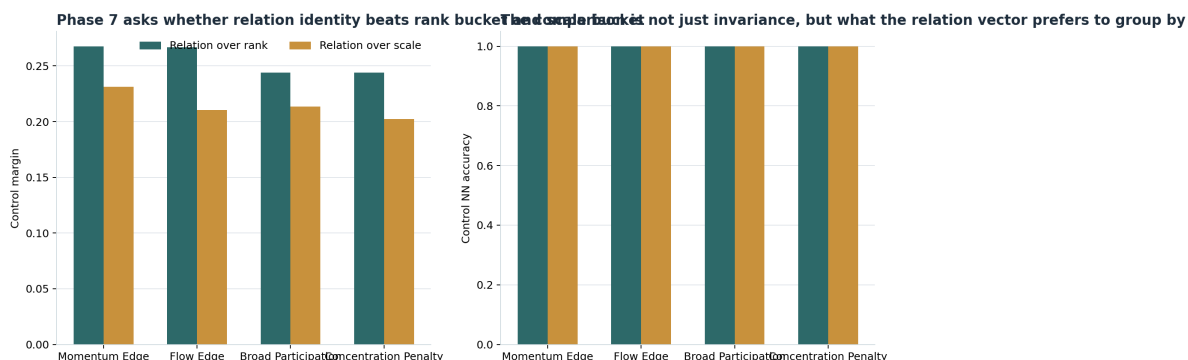
Scenario	Best Mode	Best Margin	NN Acc.
momentum edge	roster-only	0.2679	1.000
flow edge	roster-only	0.2669	1.000
broad participation	layout-only	0.2466	1.000
concentration penalty	roster-only	0.2442	1.000

Two details are most important:

- the best results all occur at row_mean @ layer 1
- every control mode still hits 1.0 nearest-neighbor accuracy in the best read

This is much stronger than the earlier row-retrieval story. The same relation survives layout, roster, and scale changes well enough that the nuisance axis no longer dominates the grouping.

Relation Beats Rank Bucket And Scale Bucket



The real question is not just whether relation retrieval works, but whether the vector prefers the same relation over confounds that share the same rank bucket or magnitude scale.

Scenario	Rank Margin	Scale Margin	Best State
momentum edge	0.2677	0.2312	row_mean L1
flow edge	0.2669	0.2104	row_mean L1
broad participation	0.2440	0.2137	row_mean L1
concentration penalty	0.2439	0.2024	row_mean L1

This is the headline result of Phase 7. Relation identity does not merely survive nuisance variation in the abstract; it specifically beats:

- matching by the same anchor rank bucket
- matching by the same global magnitude bucket

That is exactly the control the previous report called for.

Interpretation

SUPPORTED NOW

Relation-first was the right methodological move. In this synthetic family, the model's market representation groups strongly by anchor-pair relation identity even when rank context and magnitude scale are changed.

LIMITATION

The task is probably still too easy. Everything peaks at row_mean @ layer 1, which suggests Phase 7 is confirming the object shift more than revealing a deep later-stage abstraction.

So the correct reading is:

- this is a real win for the relation-first program
- but not yet the final form of the market-representation experiment

The next dataset should preserve the relation-first target while making the negatives much harder:

- relation pairs with more closely matched raw factor deltas
- confounds that share pairwise ordering but differ in which factor actually drives the edge
- harder surface/layout disturbance without letting row_mean @ L1 solve the task trivially

Conclusion

Phase 7 does exactly what the previous report asked for:

- it replaces row identity with relation identity
- it adds explicit rank and magnitude controls
- and it shows that relation identity wins those comparisons cleanly

That means the project should keep moving **upstream** and **relational**, not slide back to row retrieval or end-state labels.

Synthetic Market Phase 7 Relation Invariance — 21 March 2026. Corrected Phase 7 rerun after fixing the scenario-anchored comparison pool for the new relation-over-rank and relation-over-scale controls.